Study notes: Section 3: emf of a rotating coil; mean, r.m.s. and virtual values (Murugeshan 13.1)

Murugeshan, *Electricity and Magnetism* (10th ed.), Ch. 12 §12.1 and §12.2 (Module 4, syllabus item 15–16). Figures are drawn for these notes from the book formulas; the book figure number is in each caption. Problems are not included.

13.1 EMF induced in a coil rotating in a magnetic field

Source: book-sources/m4-s3-ac-emf.html, Murugeshan pp. 329–333 (printed), with the impedance definitions at printed p. 333. Syllabus items 17 (and the peak, mean and r.m.s. values needed for items 18–22). The book does not number the equations of this section, so they are shown unnumbered.

Before you start

- Magnetic flux $\phi = BA\cos\theta$ and Faraday's law (pre-physics P4).
- Sinusoidal functions and their integrals (pre-physics P5; pre-math M5, M7).

1. The rotating coil

Take a rectangular coil of N turns, of length a and width b, rotating with uniform angular velocity ω about its axis in a uniform magnetic field $\mathbf{B}$ (Fig. 13.1). The axis is at right angles to the field. As the coil turns, the flux through it changes, so an emf is induced.

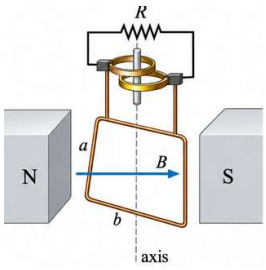


Fig. 13.1 (book): rectangular coil rotating between poles N and S, with slip rings and resistance R.

Start timing when the plane of the coil is at right angles to $\mathbf{B}$, that is when the normal to the coil makes zero angle with $\mathbf{B}$. At time t the normal makes angle $\theta = \omega t$ with $\mathbf{B}$. The flux linked with N turns is

$$\phi = NBA\cos\theta = NBA\cos\omega t, \qquad A = ab$$

2. The induced emf

By Faraday's law, the instantaneous emf is

$$E = -\frac{d\phi}{dt} = NBA\,\omega\sin\omega t = E_0\sin\omega t$$

where $E_0 = NBA\omega$ is the *peak value* of the emf. With $\omega = 2\pi\nu$, the emf is a sine curve (Fig. 13.2). A coil in a closed circuit carries an *alternating current* $I = I_0\sin\omega t$.

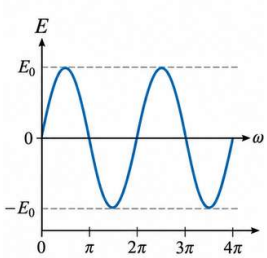


Fig. 13.2 (book): emf E = E_0 sin ωt against ωt; a sine curve.

3. One cycle

The time for one cycle is the *time period* $T = 1/\nu$. The peak value of the current or voltage is the *amplitude* (Fig. 13.3).

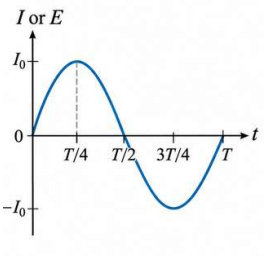


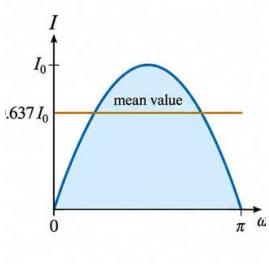
Fig. 13.3 (book): one cycle of ac voltage or current, with peak value and the times T/4, T/2, 3T/4, T.

4. Mean value

The mean value is the average over half a cycle:

$$I_{mean} = \frac{\int_0^{\pi/\omega} I_0 \sin \omega t \, dt}{\pi/\omega} = \frac{2I_0}{\pi} = 0.637\, I_0$$

Similarly $E_{mean} = 0.637\, E_0$. The mean is the area under the half-cycle curve divided by its base (Fig. 13.3 shading).



Mean value over half a cycle, 0.637 I_0 (book, §13.1, shaded area).

5. Root mean square (r.m.s.) value

The r.m.s. value is the square root of the average of I^2 over a complete cycle:

$$\overline{I^2} = \frac{\int_0^{2\pi/\omega} I_0^2 \sin^2 \omega t \, dt}{2\pi/\omega} = \frac{I_0^2}{2}$$

$$I_{rms} = \frac{I_0}{\sqrt{2}} = 0.707\, I_0$$

Similarly $E_{rms} = E_0/\sqrt{2} = 0.707\, E_0$.

6. Virtual (effective) value

The r.m.s. value is also called the *virtual* or *effective* value. It is the direct current that would produce the same heat in the same resistance in the same time. Equating the heat from $I = I_0 \sin \omega t$ to $I_v^2 R T$ over one period gives $I_v = I_0/\sqrt{2}$ again.

$$I_v = \frac{I_0}{\sqrt{2}} = I_{rms}$$

7. Form factor

The form factor measures the shape of the wave. It is the ratio of the r.m.s. value to the mean value:

$$\text{Form factor} = \frac{I_{rms}}{I_{mean}} = \frac{0.707\, E_0}{0.637\, E_0} = 1.11$$

8. Mean emf over a half cycle

With $E = E_0 \sin \omega t$, integrating over half a period $T/2$ gives $E_{av} = 2E_0/\pi$. The r.m.s. emf is obtained from the mean of E^2 over one period, using $\int_0^T \sin^2 \omega t \, dt = T/2$, which gives $E_{rms} = E_0/\sqrt{2}$.

9. Impedance, reactance and admittance

In an AC circuit the current is controlled by resistance, inductance and capacitance. The *impedance* Z is the ratio of the applied emf to the current, $Z = E/I$. It is the hindrance the circuit offers to the alternating current. The part of the hindrance due to inductance and capacitance is the *reactance*. Inductive reactance is X_L and capacitive reactance is X_C. The inverse of the impedance is the *admittance* $Y = 1/Z$.